

Learning to Simulate Cloth: Physics-Informed Deep Learning for Deformable Objects

State of the Art Report

Antonin Granados

Télécom Paris
Supervisor : Kiwon Um

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Motivation



Images from Marvel Studios, Ubisoft, Wētā FX (click image to see a demo video)

- Cloth simulation: a core problem in computer graphics
 - film and animation,
 - video games,
 - virtual reality and virtual try-on.
- Physics-based simulators:
 - high realism,
 - poor real-time performance.

Research Problem

- Key idea: learn cloth dynamics from data.
- Expected benefits of learning-based approaches:
 - faster inference,
 - easier real-time integration.
- Main question of this report:

How can we leverage physics-informed deep learning to produce realistic cloth simulations for computer graphics applications?

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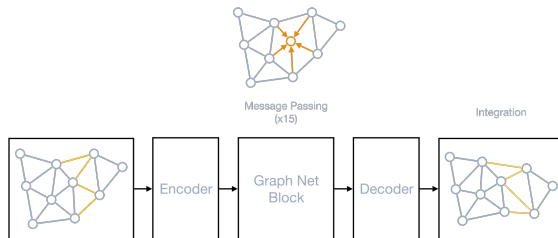
Meshes and Graph Neural Networks



(left) T-Shirt as a mesh, (right) T-Shirt rendered

- Cloth represented as **meshes**:
 - vertices define positions, edges encode constraints (stretch, bend, ...)
- Simulating cloth:
 - predict vertex positions over time,
 - each time step updates the mesh geometry.
- **GNNs** predict how each vertex moves based on its neighbors.

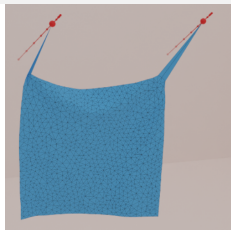
MeshGraphNets: A Key Architecture



- **MeshGraphNets (MGN)**: reference architecture for learned simulation
- Key hypothesis:
 - physical interactions are mostly local (stretching, bending, collision),
 - local message passing can model global behavior.
- What the model learns:
 - per-vertex accelerations (integrated over time),
 - replacing force computation in a classical solver.

Pfaff, Fortunato, Sanchez-Gonzalez, Battaglia. *Learning Mesh-Based Simulation with Graph Networks*. NeurIPS, 2021.

Beyond the Original MGN



(left) long-range effects and temporal instability, (center) collision failures, (right) skeletal representation

- Extensions of MGN:

- long-range interactions → unrealistic global motion artifacts,
- temporal stability → error accumulation,
- collision handling → cloth/object penetration.

- Alternative approaches:

- skeletal representations,
- neural operators.

Grigorev et al. *HOOD: Hierarchical Graphs for Generalized Modelling of Clothing Dynamics*. 2022.

Liao et al. *SENC: Handling Self-collision in Neural Cloth Simulation*. 2024.

Libao et al. *MeshGraphNetRP*. MIG, 2023.

Zhang et al. *FastClothGNN*. Graphical Models (journal), 2025.

Chen et al. *FNOPT*. 2025.

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Training data acquisition



Mesh from ARCSim simulations (left) and a VTO dataset (right)

- **Real-world** cloth motion:
 - difficult to capture accurately.
- **Synthetic** data:
 - generated using physics-based simulators (e.g., ARCSim),
 - scalable and controllable,
 - computationally expensive.

Narain, Samii, O'Brien. *Adaptive Anisotropic Remeshing for Cloth Simulation*. TOG, 2012.

Narain, Pfaff, O'Brien. *Folding and Crumpling Adaptive Sheets*. TOG, 2013.

Santesteban, Otaduy, Casas. *Learning-Based Animation of Clothing for Virtual Try-On*. CGF (Eurographics), 2019.

Learning Strategies

- **Supervised learning:**
 - learns simulator outputs,
 - accurate short-term predictions,
 - error accumulation over long rollouts.
- **Self-supervised learning:**
 - no ground-truth trajectories,
 - losses based on physical energies and constraints,
 - longer to train and harder to obtain realistic results.
- **Semi-supervised learning:**
 - combines data loss and physical constraints,
 - improves temporal stability.

Supervised: Pfaff et al. *Learning Mesh-Based Simulation with Graph Networks*. NeurIPS, 2021.

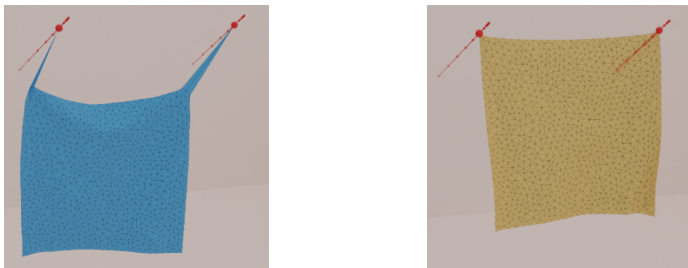
Self-supervised: Santesteban, Otaduy, Casas. *SNUG: Self-Supervised Neural Dynamic Garments*. 2022.

Semi-supervised: Libao et al. *MeshGraphNetRP*. MIG, 2023.

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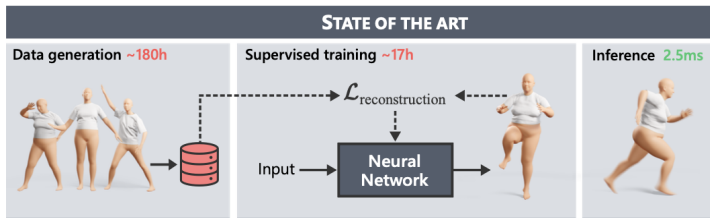
Temporal Stability and Generalization



Error accumulation in MeshGraphNets (left) - Ground truth (right)

- Most models are autoregressive $\rightarrow$ errors accumulate over time
- Generalization issues:
 - unseen materials,
 - new mesh resolutions or topologies.
- GNNs generalize better than skinning-based methods,
- but material changes often require retraining.

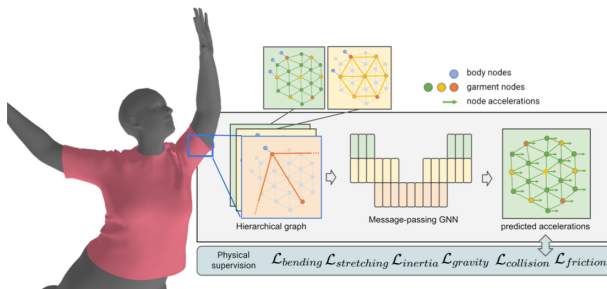
Dependence on Synthetic Data



Requirement for long and complex simulations

- Training requires large, high-quality datasets.
- Dataset generation is:
 - computationally expensive,
 - potentially biased by simulator inaccuracies.
- Physics-informed losses help,
- but data generation remains a bottleneck.

Scalability and Performance



Hierarchical architecture example

- GNN complexity grows with mesh resolution.
- High-resolution garments are difficult to handle:
 - memory limitations,
 - slow inference.
- Multiscale and hierarchical methods exist,
- but often trade accuracy for efficiency.

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Conclusion and Perspectives

- Physics-informed learning is a **promising direction** for cloth simulation
- Graph-based models achieve strong realism and show real-time potential
- However, **key challenges remain**:
 - long-term temporal stability,
 - strong dependence on classical simulators,
 - limited scalability to high-resolution meshes.
- **Directions I would like to explore**:
 - improving scalability using attention or sparse message passing,
 - reducing reliance on synthetic data through self-training.

Thank you for your attention!